

Operational risk management

12.1 Context and objectives of the analysis

Recent legislation and market practices make it important to measure the risks arising as a result of management decisions. For instance, the Basel II capital accord, published by Basel Committee on Banking Supervision (BCBS, 2001), requires financial institutions to measure operational risk, defined as ‘the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events’. Another important standard is the recently published ISO 17799 which establishes the need for risk controls aimed at preserving the security of information systems. Finally, Publicly Available Specification PAS 56, in setting criteria that need to be met in order to maintain the business continuity of IT-intensive companies, also calls for the development of statistical indicators for monitoring the quality of business controls in place. In this chapter we focus on the Basel accord, while keeping in mind that what is developed here in a banking context can be extended to the general enterprise risk management framework (see Bonafede and Giudici, 2007).

The Bank of International Settlements (BIS) is the world’s oldest financial institution; its main purpose is to encourage and facilitate cooperation among central banks (BCBS, 2001). In particular, the BIS established a commission, the Basel Committee on Banking Supervision, to formulate broad supervisory standards, guidelines and formulate best practice recommendations. The ultimate purpose of the BCBS is to prescribe capital adequacy standards for all internationally active banks.

In 1988 the BCBS issued one of the most significant international regulations impacting on the financial decision making of banks, the Basel accord. This was later replaced by the New Accord on Capital Adequacy, or Basel II (BCBS, 2001). This new framework, developed to ensure the stability and soundness of financial systems, was based on three ‘pillars’: minimum capital requirements, supervisory review and market discipline. What was important about the new accord was that it identified operational risk as a distinct category of risk. In fact, it was only with the new accord that the Risk Management Group of the Basel Committee proposed the current definition of operational risk, stated in the

opening paragraph of this section. The Risk Management Group also provided a standardised classification of operational losses into eight business lines and seven event types, giving 56 possible categories.

The aims of operational risk measurement (Alexander, 2003; King, 2001; Cruz, 2002) are twofold. On one hand, there is a prudential aspect which involves setting aside an amount of capital to cover unexpected losses. This is typically done by estimating a loss distribution and deriving functions of interest from it, such as the value at risk (VaR). On the other hand, there is a managerial aspect, for which the issue is to rank operational risks in an appropriate way, say from high priority to low priority, so to individuate appropriate management actions directed at improving preventive control of such risks. In general, the measurement of operational risks leads to the measurement of the efficacy of controls in place at a specific organisation: the higher the operational risks, the worse such controls.

The complexity of operational risks and the newness of the problem have driven international institutions, such as the BCBS, to define conditions that sound statistical methodologies should satisfy to build and measure operational risk indicators.

12.2 Exploratory data analysis

Operational risk is composed of two elements: the frequency of loss events over a given period of time (frequency) and the mean monetary impact of the loss over the given period (severity). The following sources of information are generally used to determinate the operational value at risk:

- **Historical data.** In its simplest form this is a table whose rows contain information about the loss, such as the amount, the date of occurrence, the organisational unit identifying the loss, the amount recovered by the insurance and an indication of the business line and event type.
- **Expert opinion.** This is risk evaluations made by experts (branch heads, area heads, etc.) about the activity performed by the bank.
- **External databases.** These are typically consortium databases which aggregate the operational losses of all the banks taking part in the consortium. Here we shall refer to the Data Base Italiano delle Perdite Operative (Italian Database of Operational Losses, DIPO) consortium.

It is reasonable to expect some missing values among the events that constitute the universe of possible losses. Certain events have never happened in many situations or have never been registered either internally or at DIPO level. Sometimes experts do not manage to make evaluations. Thus some categories may have all three sources of information, while others have only two, one or even no sources.

The various sources of information are supplemented by a process of scaling. This makes it possible to compare and aggregate different databases, at least in terms of analysis. For example, we expect the DIPO, being an aggregation of the losses of several banks, to have loss amounts higher than the internal loss amounts. On the other hand, experts' opinions are formulated in specific areas, so their values may be rather different and, possibly, be comparable with internal losses.

In the DIPO case, scaling is done by dividing the losses in every category by a constant which is equal to the ratio between the total DIPO losses and the total internal losses. The same method can be applied to the frequency, so that we obtain the severity as a ratio. Obviously, in order for this scaling to be effective, the bank should have the same features as DIPO; given that DIPO gathers information from a set of banks in the same country, this requirement should be satisfied.

Another data organisation activity typically performed at the beginning of operational risk modelling is mapping. This is often a preliminary to the survey of expert opinion. By mapping we mean the identification, coding and schematization of all the activities carried out by a bank. This way, the organisation can be seen as an aggregation of productive processes, which are articulated as subprocesses, phases and subphases. The structure can be more or less complex according to the number of hierarchical levels considered. At the most elementary level, connected risks are identified. We talk about a risk when, in a given activity, there is a non-zero probability of incurring a loss.

Once mapping has been done, whether in the manner described or in a simpler manner starting with an exhaustive list of risk events (regardless of the processes), the expert must describe how the risk is perceived in terms of frequency and severity. Such opinions must refer to the Basel accord business lines and events types, so that each perceived loss can be mapped to a risk category.

The analysis of expert opinion is usually the exploratory analysis that is run in operational risk measurement. For the data set we have analysed, originating from a medium-sized Italian bank, we have noticed that experts agree about the distribution of the hazardousness across the losses, but not about their absolute dimension. This may be because different business areas have various levels of operational risk, as perceived by the experts. A scale factor could solve this problem, which will thus distinguish the various evaluations. Some experts may consider a loss of €10 000 as more or less insignificant, while for others it is very important.

Once expert opinions have been gathered, they are usually analysed with simple frequency distributions and graphs, one for each category. Aggregating the risks at the category level, we will obtain a database with 56 frequency $\times$ severity entries, one for every business line and event type. The product of the frequency and the severity gives the perceived loss for each category, and the sum of all such losses gives a qualitative (perceived) estimate of the total loss of the bank for every category.

Missing data is a problem that often occurs with expert opinions. This is sometimes due to the lack of evaluations by a few experts, sometimes due to the lack

of a complete mapping activity of the process, which prevents the identification of some risks. In our case only two area experts have been asked. The mapping activity of the process, and the consequent identification of the risks, is exclusively concerned with the payment and settlements business area. To bypass the first missing-data problem we used the dimension of a business area (number of area branches related to the total number) to assess the quality of the data at our disposal. Therefore, for instance, since the payment and settlements area represents about the 17% of the total losses, the estimate of the total value on the basis of expert opinions has been obtained by dividing this value by 0.17. The idea to weight the evaluations by the number of branches is equivalent to the hypothesis that, at the branch level, the risk is almost identical, so that the risk of a broader area depends on its number of branches.

Another way to weight and scale opinions is to use key risk indicators (KRIs), variables which have a certain association with the losses and act as ‘indicators’ of the risk. Such variables are identified during the process mapping and monitored throughout the year. We decided to correct our experts’ opinions using the KRIs available. The size of the correction depends on the significance of the KRI. For this purpose every loss, pertaining to the i th risk of the j th category evaluated by the k th expert, is multiplied by a factor given by

$$F_{i,j,k} = \begin{cases} 1 + 0.3R_{i,j}, & \text{if } \bar{K}_{i,j,k} > 0.66, \\ 1, & \text{if } 0.33 < \bar{K}_{i,j,k} < 0.66, \\ 1 - 0.3R_{i,j}, & \text{if } \bar{K}_{i,j,k} < 0.33, \end{cases}$$

where $\bar{K}_{i,j,k}$ is the average KRI associated with the i th risk of the j th category calculated on branches belonging to the area of the k th expert; and $R_{i,j}$ is the ratio between the deviation of the means of the area and total deviation relative to the i th risk and to the j th category.

The R^2 index represents the share of deviance that the classification is able to explain in terms of areas. This way, the amount of the correction is directly proportional to the significance of the KRI. If a KRI is not very discriminating from area to area, R^2 is small and so is the correction size. Conversely, if the KRI is appreciably different from area to area, the F factor will be larger.

12.3 Model building

Statistical models for operational risk can be grouped into two main categories: ‘top-down’ and ‘bottom-up’ methods. In top-down methods, risk estimation is based on macro data without identifying the individual events or the causes of losses. Operational risks are measured and covered centrally, so local business units are not involved in the measurement and allocation process. Top-down methods include the Basic Indicator Approach (Yasuda, 2003; Pézier, 2002) and the Standardised Approach (Cornalba and Giudici, 2004; Pézier, 2002), where risk is computed as a certain percentage of the variation of some variable, such as gross income, considered as a proxy for company performance. This approach is

suitable for small banks, which prefer a methodology that is cheap to implement and easy to implement.

Bottom-up techniques, on the other hand, use individual events to determine the source and amount of operational risk. Operational losses can be divided into levels corresponding to business lines and event types, and the risks are measured at each level and then aggregated. These techniques are particularly appropriate for large banks and those operating at the international level, since they can afford the implementation of sophisticated methods which sensitive to the bank's risk profile. Advanced Measurement Approaches (AMA) belong to this class (BCBS, 2001). Under the AMA, the regulatory capital requirement will equal the risk measure generated by the bank's internal operational risk measurement system using the quantitative and qualitative criteria set by the BCBS. This is an advanced approach as it allows banks to use external and internal loss data as well as internal expertise (Giudici and Bilotta, 2004).

Statistical methods for operational risk management in the bottom-up context have recently been developed. One main approach has emerged: the actuarial approach. This method uses actual loss data to estimate the probability distribution of the losses. The most popular methods (King, 2001; Cruz, 2002; Frachot *et al.*, 2001; Dalla Valle *et al.*, 2008) are based on extreme value distributions. Another line of research suggests the use of Bayesian models (Yasuda, 2003, Cornalba and Giudici, 2004; Fanoni *et al.*, 2005; Dalla Valle and Giudici, 2008). The main disadvantage of actuarial methods is their backward-looking perspective; their estimates are based entirely on past data. Furthermore, it is often the case, especially for smaller organisations, that for some business units there are no loss data at all. Regulators thus recommend developing models that can take into account different data streams, not only internal loss data (see BCBS, 2001). These streams may be: self assessment opinions, usually forward looking; external loss databases, usually gathered through consortia of companies; and data on key performance indicators.

In the actuarial model, loss events are assumed independent and, for each one, it is assumed that the total loss in a given period (e.g. one year) is obtained as the sum of a random number (N) of impacts (X_i). In other words, for the j th event the loss is

$$L_j = \sum_{i=1}^{N_j} X_{ij}.$$

Usually the distribution of each j -specific loss is obtained from the specification of the distribution of the frequency N and the mean loss or severity S . The convolution of the two distributions leads to the distribution of L (typically via a Monte Carlo estimation step), from which a function of interest, such as the 99.9 percentile (the value at risk) can be derived.

The scorecard approach is based on the so-called self assessment, which is based on the experience and the opinions of a number of internal 'experts' of the company, who usually correspond to a particular business unit. An internal procedure of control self assessment can periodically be done by means of

questionnaires. Questionnaires can be submitted to risk managers (experts), and give information such as the quality of internal and external control systems of the organisation based on their experience over a given period. In a more sophisticated version, experts can also assess the frequency and mean severity of the losses for such operational risks (usually in a qualitative way). Self assessment opinions can be summarised and modelled so as to obtain a ranking of the different risks, and a priority list of interventions in terms of improvement of the related controls.

In order to derive a summary measure of operational risk, perceived losses contained in the self-assessment questionnaire can be represented graphically (e.g. in a histogram), leading to an empirical non-parametric distribution. Such a distribution can be employed to derive a function of interest, such as the VaR.

Scorecard models are useful for prioritising interventions in the control system, so as to effectively reduce the impact of risks, *ex ante* and not *a posteriori*, by allocating capital for example (corresponding to the VaR).

In Giudici (2008) a methodology aimed at summarising concisely and effectively the results of a self-assessment questionnaire has been proposed. In the next section we shall show how it can be applied.

12.4 Model comparison

Suppose that we are given 80 events at risk (this is the order of magnitude employed in a typical operational risk management analysis in the banking sector). The events can be traced to the four main causes of operational risk: People, processes, systems and external events. First of all a sample of banking professionals (from headquarters and the local branches) is obtained. The aims of the questionnaire project are described in a group presentation. The nature and structure of each risk question will have been devised in a focus group discussion with the top manager of the bank. The result of this preliminary analysis is that each of the professionals is asked his/her opinion on the frequency, severity and effectiveness of the controls in place for each risk event. The number of possible frequency classes is equal to four: daily, weekly, monthly, and yearly. The number of severity classes depends on the size of the bank's capital, and will typically be six or seven, ranging from 'an irrelevant loss' to 'a catastrophic loss'. Finally there are three possible classes for the controls: not effective, to be adjusted, and effective.

Once the interviews have been gathered together, the aim is to assign a 'rating' to each risk event, based on the distribution of the opinions on the frequency, controls and severity. Giudici (2008) suggests using the median class as a location measure for each distribution, and the normalised Gini index as an indicator of the 'consensus' on such location measure. This results in three rating measures for each event, denoted by the conventional risk letters: A for low risk, B for medium risk, C for higher risk and so on. While the median is used to assign a single-letter measure, the Gini index is used to double or triple the letter, depending on the value of the index. For example, if the median of the frequency distribution of

a certain risk type (e.g. theft and robbery) is yearly, corresponding to the lowest risk category, a letter A is assigned. Then, if all the interviewees agree on that evaluation (e.g. the Gini index is equal to zero), A becomes AAA; if the Gini index corresponds to maximum heterogeneity, then A remains A. Intermediate cases will receive an AA rating. The same approach will serve for the severity as well as for the controls, leading to a complete scorecard that can be used for intervention purposes. For visualisation purposes, colours are associated with letters, using a traffic-light convention: green (grey); corresponds to A; yellow (light grey) to B; red (dark grey) to C and so on.

Figure 12.1 presents the results from such scorecard model for a collection of risk events caused by to people (internal fraud) and external events (external fraud and losses from material activities). It turns out that event 1.2.6 should be given top priority for intervention, as controls are not effective, and both frequency and severity are yellow. Other events at risk include 2.2.1 and 2.2.4, which have a high frequency and medium quality controls. We observe that opinion on the severity is usually considered second in priority determination as it typically concerns a mean value which cannot be modified by the action of controls.

While scorecard methods typically use self-assessment data, actuarial models use internal loss data. The disadvantage of these approaches is that they consider only one part of the statistical information available to estimate operational risks. Actuarial methods rely only on past loss data and therefore do not consider important information on the perspective and the evolution of the company; on the other hand, scorecard methods are based only on perceived data (are forward-looking) and therefore do not reflect past experience very well.

A further problem is that, especially for rare events, a third data stream may be considered: external loss data. This source of data is made up of pooled records of losses, typically higher than a certain value (e.g. €5000), collected by an appropriate association of banks.

It thus becomes necessary to develop a statistical methodology that is able to merge three different data streams in an appropriate way, while maintaining simplicity of interpretation and predictive power. Here we shall propose a flexible non-parametric approach that can achieve this objective (see Giudici, 2008). Such an approach can be justified within a non-parametric Bayesian context.

Our approach considers, for each event, data on all losses that have occurred in the past as well as the expected self assessment losses for the next period. The latter is counted as one data point, typically higher than actual losses, even when calculated as a mean loss rather than as a worst case loss. Putting together the self-assessment data point with the actual loss data points, we obtain an integrated loss distribution, from which a VaR can be calculated. Alternatively, to take the loss distributions more correctly into account, a Monte Carlo simulation can be based on the given losses, leading to a (typically higher) Monte Carlo VaR, parallel to what is usually done in the actuarial approach.

In Figure 12.2 we compare, for a real database, the VaR obtained using a pure self-assessment approach, with the actuarial VaRs (both historical and Monte Carlo based) and the integrated (Bayesian) VaR (both simple and Monte Carlo). For reasons of predictive accuracy, we construct all methods on a series of data

PEOPLE		CONTROLS	FREQUENCY	SEVERITY
Internal fraud	1.1.1	Transactions not reported (intentional)		
	1.1.2	Trans type unauthorised (w/ monetary loss)		
	1.2.1	Fraud/credit fraud/worthless deposits		
	1.2.2	Theft/extortion/embezzlement/robbery		
	1.2.3	Malicious destruction of assets		
	1.2.4	Forgery		
	1.2.5	Check kiting or smuggling		
	1.2.6	Account take-over/impersonation/etc.		
	1.2.7	Tax non-compliance/evasion (wilful)		
	1.2.8	Bribe/kickbacks		
	1.2.9	Insider trading (not on firm's account)		
	1.2.10	Theft of information (w/ monetary loss)		
	2.1.2	Hacking damage		
	2.2.1	Theft/Robbery		
EXTERNAL	2.2.2	Forgery		
	2.2.3	Check kiting		
	2.2.4	Clonation of credit cards, p.o.s., atm		
	5.1.1	Natural disaster losses		
Danni ad attività materiali	5.1.2	Losses from external sources (terrorism, vandalism)		

Figure 12.1 Example of results from our proposed scorecard model.

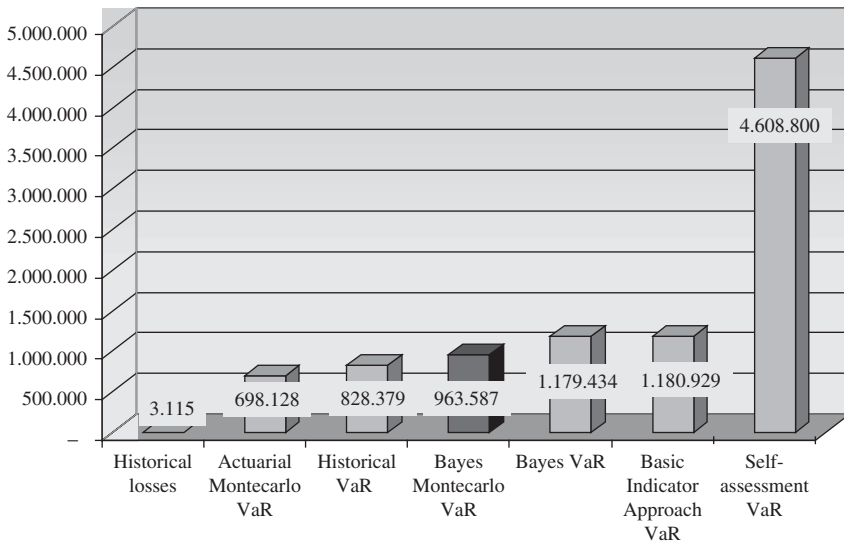


Figure 12.2 Example of results from our integrated scorecard model.

points updated at the end of 2005; calculate the VaR for 2006 (possibly integrating it with the self assessment available opinions for 2006) and compare the VaR with the actual losses for 2006. We also calculate the VaR that would be obtained under the simple Basic Indicator Approach (BIA) suggested by Basel II. The BIA amounts to calculating a flat percentage (15%) of a relevant indicator (such as the gross income), without further elaboration. It turns out that both our proposed models (Bayes VaR and Bayes Monte Carlo) lead to an allocation of capital (represented by the VaR) lower than the BIA and higher than the observed losses. Although these results are achieved by the actuarial models as well (historical and actuarial Monte Carlo), we believe that a prudential approach, as expressed by our proposal, is more sound, especially over a longer time horizon.

12.5 Summary conclusions

1. **Context.** We have shown how to develop efficient statistical methods for measuring the performance of business controls, through the development of appropriate operational risk indicators.
2. **Objectives.** The advantages of measuring operational risk appropriately are twofold. On the one hand, there is a prudential aspect which involves setting aside an amount of capital that can cover unexpected losses. On the other hand, there is a managerial aspect for which the issue is to rank operational risks in an appropriate way, say from high priority to low priority, so as to identify appropriate management actions directed at improving preventive controls on such risks.

3. **Organisation of the data.** The data analysed in operational risk usually comes from three sources: expert opinion, often corrected by means of key risk indicators; internal loss data; and external loss data. We have shown how to scale and integrate these databases together.
4. **Exploratory data analysis.** The exploratory data analysis is based on first merging the databases available by means of scaling techniques. Simple frequency distributions and graphs are usually obtained.
5. **Model specification.** Different statistical models can be developed, compared and then used, depending on the available databases. The main modelling strategies are: actuarial, scorecard and Bayesian.
6. **Model comparison.** Modelling strategies have been compared not only theoretically but also with reference to an available database, according to the data mining paradigm. The main comparison benchmark is predictive performance (known in this context as backtesting).
7. **Model interpretation.** The measurement of operational risk has been made compulsory by recent legislative changes. We have shown that, when models are transparent and interpretable, the measurement of operational risk leads to the measurement of the efficacy of controls in place at a given organisation: the higher the operational risks, the worse the controls.